Amirhossein Eskorouchi

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Project Portfolio

Summary

Machine Learning and Optimization Researcher with 4+ years of experience developing novel AI methods and intelligent decision systems across manufacturing, healthcare, and engineering applications. Specializes in computer vision and optimization-guided learning, integrating domain knowledge, physical constraints, and mathematical optimization to develop reliable, interpretable, and decision-relevant AI systems. Proven ability to lead end-to-end R&D, including problem formulation, algorithm design, large-scale experimentation, rigorous validation, and deployment-oriented prototyping.

Professional Experience

Machine Learning & Optimization Researcher **Jan 2024 – Present**
Systems Intelligence and Optimization Lab, Mississippi State University

- Developed **UOT-DETR**, integrating **unbalanced optimal transport** into PyTorch/DETR for small-object detection and size estimation; reduced distribution error by up to **30%** and increased recall by **24.7 percentage points** with no inference-time overhead.
- Developed a **multi-horizon power-outage forecasting model** combining **CNN/LSTM networks and unbalanced optimal transport**; **Finalist, 2026 INFORMS Data Mining Society Data Challenge**.
- Developed **AnatoECE**, a **knowledge-informed 3D transformer** using anatomical constraints to guide segmentation and diagnostic predictions; combined **self-supervised pretraining on 1,500 unlabeled CT volumes** with limited labeled data, achieving **0.928 AUC** and **0.823 Dice** for cancer assessment.
- Engineered **Wood-Chip Monitor**, a portable **NVIDIA Jetson/TensorRT edge-AI system** for real-time detection, size measurement, moisture classification, and quality monitoring in biomass manufacturing.
- Built a **graph-based multimodal ML pipeline** analyzing **40K+ sleep-study epochs across 17 physiological channels**; achieved **85% recall** for ADHD identification and identified sleep-stage-specific biomarkers.
- Co-developed **PlantLife365**, a multisensor **IoT and ML platform** combining predictive analytics and an **LLM-based assistant** for real-time agricultural monitoring and decision support.

M.Sc. Researcher **Sep 2021 – Jun 2023**
Iran University of Science and Technology, Tehran, Iran

- Developed decision models combining **Dempster–Shafer evidence theory, robust optimization, and scenario analysis** for portfolio, energy, and market decisions under uncertainty.

Technical Skills

Programming & Libraries: Python, SQL, R, MATLAB; PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy
Machine Learning & Computer Vision: Deep Learning; Self-Supervised and Knowledge-Informed Learning; Transfer Learning and Domain Adaptation; Graph Learning; Object Detection; 2D/3D Segmentation
Optimization & Decision Science: Linear and Nonlinear Programming; Robust and Multi-Objective Optimization; Optimal Transport; Gurobi, CPLEX
Data Science & Analytics: Predictive Modeling; Time-Series Forecasting; Feature Engineering; Statistical Inference; Design of Experiments; Data Visualization and Power BI
AI Engineering & Deployment: End-to-End ML Pipelines; Model Validation; Edge AI and Real-Time Inference; NVIDIA Jetson, TensorRT; FastAPI, Git

Education

Ph.D.: Industrial and Systems Engineering, Mississippi State University, MS, USA (GPA: 4.00/4.00) **2024 - Present**
Dissertation: "Evidence-Constrained and Optimization-Guided Learning for Reliable Computer Vision and AI Systems"

M.S.: Industrial Engineering, Iran University of Science and Technology, Tehran, Iran (GPA: 3.52/4.00) **2018 - 2021**
Thesis: "Portfolio Optimization under Uncertainty Using Dempster–Shafer Evidence Theory"

Selected Publications & Manuscripts

- Distributionally Constrained Visual Learning with Unbalanced Optimal Transport. *Knowledge-based Systems*, under review.
- Knowledge Transferability in Differentiable Architecture Search for Skin Lesion Diagnosis. *IISE Transactions on Healthcare Systems Engineering*, 2026.
- Dempster–Shafer Energy Portfolio Optimization under Deep Uncertainty. *Results in Engineering*, 2025.

Awards, Leadership & Professional Service

- **Finalist**, INFORMS Data Mining Society Data Challenge, INFORMS Annual Meeting 2026
- **1st Place**, IISE DAIS Student Mobile/Web App Competition, *PlantLife365* 2026
- **Finalist**, IISE DAIS Student Mobile/Web App Competition, *Wood-Chip Monitor* 2026
- **3rd Place**, IISE DAIS Data Analytics Competition 2025
- **3rd Place**, Campus Vision Challenge, Mississippi State University AI Club 2025
- **Session Chair**, "Advanced AI Solutions in Health and Diagnostics," IISE Annual Conference & Expo 2025
- **President**, INFORMS Student Chapter, Mississippi State University (*Vice President, 2025–2026*) 2026–Present
- **Mentored 3 graduate and 5 undergraduate researchers** in machine learning and AI
- **Peer Reviewer**, *Engineering Applications of Artificial Intelligence*, *Computers & Operations Research*, *Journal of Cleaner Production*, and *Optics & Laser Technology*.